

MySuperstore Order Handling and Fulfillment Infrastructure Report

Prepared for MONARCH GROUP

Date	April 27, 2026
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Project	MySuperstore
Company	MONARCH GROUP
Subject	Infrastructure plan for post-purchase order handling and fulfillment
Scope	Customer order visibility, vendor fulfillment, admin operations, shipment tracking, and operational monitoring
Exclusion	Cancellation, refund, and return handling are intentionally excluded from this report

This document assumes the MySuperstore website already supports order creation, payment completion, address selection, and cart cleanup through Supabase-managed application logic.

1. Executive Summary

The MySuperstore website appears to have the order creation layer in place through a Supabase-based storefront. The main gap is the operational layer that moves a paid order through vendor fulfillment, shipment visibility, and administrative oversight. This report sets out the infrastructure required to complete that part of the platform for MONARCH GROUP.

The recommended approach is to use Supabase as the single source of truth for order status, fulfillment activity, shipment records, and operational notifications. The target outcome is a simple, auditable order pipeline for MySuperstore that customers can trust, vendors can act on, and MONARCH GROUP can supervise.

2. Project Context

- Project website: MySuperstore
- Company receiving this report: MONARCH GROUP
- Prepared by: AKACHUKWU AUGUSTINE OGBO
- Primary backend assumption: Supabase

3. Scope and Assumptions

- Order creation, payment completion, address selection, and cart clearing are assumed to be already working.
- Supabase is assumed to be the primary backend platform, including Postgres, Auth, Storage, Realtime, Edge Functions, and scheduled jobs.
- The immediate goal is to complete post-purchase order handling and fulfillment operations for MySuperstore rather than redesign the storefront itself.
- Cancellation, refund, and return handling are intentionally out of scope for this report.

4. Target Operating Model

MySuperstore should treat every paid order as a controlled workflow. Customers should see clear progress, vendors should see only the work assigned to them, and MONARCH GROUP administrators should monitor the whole process from one operational view.

- Customer view: sees status, shipment progress, vendor split details where applicable, and proof that the order is advancing.
- Vendor view: sees only assigned order items, required next action, shipping details, and service-level expectations.
- Admin view: sees the whole order, vendor participation, payment state, operational exceptions, and audit history.
- Platform view: every transition is recorded, validated, timestamped, and attributable to a user, vendor, or automated function.

5. Core Infrastructure Modules

4.1 Order Lifecycle Control

Order progression should not rely on manual status edits alone. Supabase should hold a clear lifecycle with controlled transitions such as paid, processing, packed, shipped, delivered, and completed.

- Central order status registry with allowed transitions.
- Per-vendor fulfillment status for multi-vendor orders.
- Per-item operational fields where vendors can fulfill only their own lines.
- Timestamp capture for every lifecycle milestone.
- Automated validation to prevent invalid or skipped transitions.

4.2 Fulfillment Work Queue

Vendors need an operational queue rather than a read-only order list. That queue should show the next required action for each order item assigned to that vendor.

- Vendor-assigned order queue driven by database views or materialized views.
- Priority markers such as oldest order first, due soon, or overdue.
- Vendor-only action permissions enforced through row-level security.
- Operational filters for status, shipment readiness, and vendor performance.

4.3 Shipment and Tracking Registry

Shipment information should be stored as structured data rather than free text. Once an order is packed, the platform should support carrier name, tracking number, ship date, and delivery estimate.

- Dedicated shipment records linked to orders, vendors, and where needed individual order items.
- Tracking metadata storage with validation rules and uniqueness checks.
- Optional carrier integration layer for future automated tracking updates.
- Customer-facing shipment milestones sourced from the same shipment records used by vendors and admins.

4.4 Eventing and Notifications

Order handling should be event-driven. Each important change should create a structured event that supports notifications, dashboards, and exception monitoring.

- Supabase event log table for operational events.
- Edge Functions triggered by status transitions to write notifications or downstream records.
- Realtime channels or notification tables scoped by customer, vendor, and admin audience.
- Scheduled reconciliation jobs to catch missed transitions or stale orders.

4.5 Admin Operations Layer

Administrators should have more than a simple status editor. They need an operations layer that supports review, intervention, and escalation while keeping a clean audit trail.

- Full order timeline view including payment reference, vendor participation, shipment milestones, and transition history.
- Administrative override path with reason capture when intervention is necessary.

- Operational notes and internal comments separated from customer-facing content.
- Exception queue for orders stalled beyond service thresholds.

4.6 Observability and Audit Trail

MONARCH GROUP should be able to answer three questions at any time: what state is a MySuperstore order in, who changed it, and what is delayed. That requires audit history, event logging, and simple monitoring views.

- Immutable order history records for every status change.
- Actor metadata for admin, vendor, automation, or system transitions.
- Monitoring for stuck orders, missing tracking data, and overdue vendor actions.
- Reporting views for daily order throughput, shipment aging, and vendor fulfillment performance.

6. Supabase Infrastructure Blueprint

The recommended implementation should stay as close to Supabase-native services as possible. That keeps the setup simpler, cheaper to manage, and easier to audit.

1. Use Postgres tables and relational constraints as the authoritative store for orders, order items, fulfillment assignments, shipment records, order history, and notification records.
2. Use row-level security so customers only see their own orders, vendors only see assigned order items and shipments, and admins see the full operational picture.
3. Use Edge Functions for controlled lifecycle transitions, shipment creation, vendor action validation, and event emission.
4. Use Supabase Realtime or scoped notification polling for customer and vendor updates that must appear quickly.
5. Use scheduled jobs for reconciliation tasks such as overdue fulfillment checks, missing tracking reminders, and status consistency audits.
6. Use Supabase Storage only where operational artifacts are needed, such as proof of shipment documents, courier labels, or delivery confirmation uploads.

7. Information Required From MONARCH GROUP

Before implementation is finalized, MONARCH GROUP should confirm the operating and infrastructure decisions below. These answers will shape the final fulfillment design, access rules, monitoring setup, and delivery model for MySuperstore.

- How many admin roles will manage MySuperstore orders, and what permissions should each role have?
- Will vendors fulfill orders manually at first, or is there an external warehouse or logistics partner involved?
- Will shipments be tracked manually at launch, or should the platform integrate with a courier or shipping aggregator?
- What service-level targets should vendors follow for acknowledgment, packing, and shipping?
- Will MySuperstore support multi-vendor orders from day one, or only single-vendor orders at launch?
- Should MONARCH GROUP receive daily operational reports or only exception alerts for delayed orders?
- Will vendors upload proof of shipment or delivery documents, and if so, what file retention policy is required?
- Are there any country-specific compliance or data-retention requirements that affect order history or shipment records?

8. Implementation Plan

Phase 1: Foundation and Data Control

- Finalize the order lifecycle definition and allowed status transitions.
- Create or normalize operational data structures for fulfillment and shipment management.
- Define row-level security boundaries for customer, vendor, and admin order access.
- Introduce an order history log and event log for all transitions.

Phase 2: Vendor Fulfillment Enablement

- Build the vendor work queue model and its required filters.
- Add controlled vendor actions for acknowledge, process, pack, ship, and deliver.
- Create shipment record capture, including tracking fields and courier metadata.
- Introduce alerts for orders that remain untouched beyond the target response window.

Phase 3: Customer Order Visibility

- Expose richer customer-facing order milestones using the same backend status model.
- Show shipment metadata and tracking progress clearly in the account experience.
- Support multi-vendor order presentation so customers understand split fulfillment.
- Ensure every visible status is sourced from authoritative operational records rather than duplicated frontend logic.

Phase 4: Admin Control and Monitoring

- Expand the admin order detail view into an operational dashboard.
- Add exception queues for delayed acknowledgment, delayed shipping, and missing tracking.
- Add internal notes, escalation reasons, and intervention logging.
- Publish management reporting views for throughput, aging, and vendor responsiveness.

9. Key Dependencies

- A finalized marketplace order model that supports per-vendor and potentially per-item fulfillment states.
- Reliable mapping between payments, orders, vendors, and shipment records.
- Clear admin and vendor role definitions in Supabase Auth and policy design.
- Courier/tracking strategy, whether manual first or integrated later.
- Operational ownership within MONARCH GROUP for monitoring, vendor support, and escalation handling.

10. Major Risks if This Layer Is Not Implemented

- Paid orders may exist without a disciplined fulfillment path, creating manual confusion.
- Vendors may see orders but still lack a clean operational way to act on them.
- Customers may lose confidence if post-purchase visibility is weak or inconsistent.
- Admins may rely on manual status edits without a trustworthy audit trail.
- Multi-vendor orders may become difficult to coordinate once order volume increases.

11. Success Criteria

- Every paid order enters a controlled lifecycle automatically.
- Every vendor can act only on the portions of the order assigned to that vendor.
- Every shipment has structured metadata and a visible fulfillment trail.
- Customers can see trustworthy progress from paid through delivered.
- Admins can identify stalled orders, intervene when needed, and audit what happened afterward.

12. Recommended Immediate Next Steps

7. Approve the lifecycle state model and the marketplace fulfillment ownership model.
8. Provide the outstanding infrastructure and operating decisions listed in this report.
9. Define the minimum shipment record fields and the vendor action set required for first release.
10. Implement the order history log and operational event log before building additional UI controls.
11. Stand up the vendor fulfillment queue and admin exception queue as the first operational surfaces.
12. Run an end-to-end pilot using test orders across customer, vendor, and admin roles before public launch.

13. Closing Note

MySuperstore appears close to being commercially usable from an order creation standpoint. The next step for MONARCH GROUP is to formalize what happens after payment so that fulfillment becomes structured, visible, and reliable. Once MONARCH GROUP confirms the remaining infrastructure and operating assumptions, this plan can be converted into a detailed implementation workstream.